

Biomass Energy

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Biomass energy encompasses a wide variety of energy technologies that use renewable plant matter, or phytomass, derived from photosynthesis as a feedstock to produce solid, liquid, and gaseous biofuels or used directly as an energy source producing heat and electricity. Biorefinery concepts are being developed that could result in the production of multiple energy carriers, as well as bioproducts from the same biomass resource.

3.1 Biomass Feedstock Technologies

3.1.1 Photosynthesis

Biomass fuels are derived from green plants, which capture solar energy and store it as chemical energy through the **photosynthetic** reduction of atmospheric carbon dioxide. Plant leaves are biological solar collectors while the stems, branches, and roots are the equivalent of batteries storing energy-rich complex carbon compounds. Elemental analysis shows that wood and grasses are about 50% carbon. The average photosynthetic efficiency of converting solar energy into energy stored in organic carbon compounds on an annual basis varies from less than 0.5% in temperate and tropical grasslands to about 1.5% in moist tropical forests (Cralle and Vietor 1989). Although seemingly quite low, the worldwide annual storage of photosynthetic energy in terrestrial biomass is huge, representing approximately 10 times world annual use of energy (Hall et al. 1993). This annual energy storage reflects the diversity and adaptability of terrestrial plants in many different climate zones, from the polar regions to the tropics.

3.1.2 Biomass Residue Resources

The majority of biomass energy used today is derived from residues associated with the production of timber and food crops in the field and the forest, as well as in their processing into final products. On the average, for each tonne of grain or timber harvested, there is an equivalent amount of stalk, straw, or

forest residues such as branch materials. Residues from agricultural crops, such as cereal straws, are already used for bioenergy in many parts of the world and represent a large, immediately accessible resource for bioenergy in the United States. Agriculture residue recovery in the U.S. will have the most potential in high-yield cropland areas, especially those in which no-till management systems are used. Under such conditions, some portion of the crop residue may be sustainably and economically recoverable for bioenergy.

Under conventional management practices, corn stover, wheat straw, and other crop residues often have greater economic value because they are left on the land to restore nutrients, reduce erosion, and stabilize soil structure. Sustainably recoverable agricultural crop residues in the U.S. are estimated to be less than 136 Mt (Walsh et al. 2000; Gallagher et al. 2003) (all biomass amounts are on a dry basis). Crop residues worldwide are estimated to have an energy value of 12.5 EJ (Hall et al. 1993).

Recoverable wood residues from all sources in the U.S. are estimated to be approximately 94.3 Mt (McKeever 2003). This includes 67.6 Mt of logging residues; 16.4 Mt of construction and demolition wastes; and 8.7 Mt of recoverable solid wood from the postconsumer urban residue stream (or municipal solid waste stream). Primary timber mills produce about 73.7 Mt of residues; however, about 97% of this is already utilized.

A potentially large wood residue resource consists of the small-diameter growing trees and woody debris that need to be removed from U.S. forests to reduce hazardous fuel loading in forest fire mitigation. Small-diameter wood is already harvested for fuelwood in many parts of the world. The estimated forest residues, worldwide have an energy value of 13.5 EJ (Hall et al. 1993). Forest industry processing residues (e.g., sawdust or black liquor from pulping processes) are already used for energy in most cases.

3.1.3 Potential Forestry Biomass Resources Worldwide

The amount of harvestable woody biomass produced by natural forests on an annual basis ranges from about 2 to 6 t ha⁻¹ y⁻¹ (with the higher yields usually in tropical regions); this could be increased to 4–12 t ha⁻¹ y⁻¹ if brought under active management. Such management would include optimizing harvesting strategies for production, fertilization, and replanting natural stands with faster-growing species. As of 1990, 10% of world forests, or 355 Mha, were actively managed. If managed forests were increased to 20%, and if 20% of the harvested material were used for energy, the annual worldwide resource of wood for energy from currently forested land would amount to between 284 and 852 Mt of available wood, or about 5.6–17 EJ of primary energy based on potential yield ranges of managed forests. The most optimistic estimates of biomass potential would provide a total primary energy resource of about 30 EJ from managed forests (Sampson et al. 1993).

3.1.4 Potential Energy Crop Production

Significantly increasing the world's biomass energy resources will require the production of high-yield crops dedicated to conversion to bioenergy. The most environmentally beneficial dedicated crop production systems will be the production of perennial plants, using genetically superior materials, established on previously cropped land, and managed as agricultural crops. Perennials such as annually harvested grasses or short rotation trees harvested on a cycle of 3–10 years minimize soil disturbance, increase the build-up of soil carbon, provide wildlife habitat, and generally require fewer inputs of chemicals and water for a given level of production.

Energy crop yields will be highest in locations where genetically superior material is planted on land with plenty of access to sunshine and water. In the moist tropics and subtropics, as well as in irrigated desert regions, yields in the range of 20–30 t ha⁻¹ y⁻¹ are achievable with grasses and trees (e.g., eucalypts and tropical grasses, and hybrid poplars in the irrigated Pacific Northwest of the U.S.). Without irrigation, temperate woody crops would be expected to yield 9–13 t ha⁻¹ y⁻¹ of harvestable biomass, though several experimental trials have demonstrated yields of improved genotypes of poplars

in the range of $12\text{--}20\text{ t ha}^{-1}\text{ y}^{-1}$ and best yields up to $27\text{ t ha}^{-1}\text{ y}^{-1}$ (Wright 1994). Unimproved temperate perennial grasses with no irrigation typically yield about $7\text{--}11\text{ t ha}^{-1}\text{ y}^{-1}$; selected grasses in experimental trials have commonly achieved yields of $11\text{--}22\text{ t ha}^{-1}\text{ y}^{-1}$ with occasional yields of $16\text{--}27\text{ t ha}^{-1}\text{ y}^{-1}$ in some years. Although grass yields vary from year to year, the Alamo variety of switchgrass has averaged around $23\text{ t ha}^{-1}\text{ y}^{-1}$ (10 dry ton/acre/year) over 12 years in the southeastern U.S., where a long growing season favors warm-season grasses (McLaughlin and Kszos personal communication).

Temperate shelterbelts and tropical agroforestry plantings could also contribute greatly to biomass energy resources. Assuming a range of possible yields, the primary energy resource potential of converting 10%–15% of cropland to energy crops has been estimated to range from a low of 18 EJ to a high of 49 EJ. A further 25–100 EJ has been estimated to be available from the conversion of grasslands and degraded areas to the production of biomass energy resources worldwide (Sampson et al. 1993).

3.1.5 Terrestrial and Social Limitations

M.J.R. Cannell provides estimates of theoretical, realistic, and conservative/achievable capacity of biofuel plantations of trees or annual crops to produce primary energy and offset global carbon emissions between 2050 and 2100 (Cannell 2003). The theoretical estimate of biomass energy potential assumes 600–800 Mha, or a maximum of 55% of current cropland area (although including large areas of previous crop land that are now degraded) and average yields increasing from 10 to $25\text{ t ha}^{-1}\text{ y}^{-1}$ providing 150–300 EJ. The realistic estimate of biomass energy potential assumes 200–400 Mha (14%–28% of current cropland area) and average yields of $10\text{ t ha}^{-1}\text{ y}^{-1}$ (37–74 EJ); the conservative/achievable estimate assumes 50–200 Mha of current cropland and average yields of $10\text{ t ha}^{-1}\text{ y}^{-1}$ (9–37 EJ).

Cannell's conservative estimate recognizes that developing countries continue to face food and water shortages and increasing populations, and that substantial economic and policy obstacles must be overcome. Additionally, the life cycle environmental and social impacts are not always strongly positive, depending on the crop management systems used and the conversion process employed.

3.1.6 Biomass Facility Supply Considerations

The location of large biorefinery facilities will be limited by local biomass supply availability and price considerations. Siting opportunities for facilities in the U.S. that require 500 t d^{-1} or less of biomass delivered at prices of $40\text{ \$ t}^{-1}$ or greater will be abundant, especially if dedicated crops are included in the supply mix. U.S. locations with abundant supplies of biomass delivered at $20\text{ \$ t}^{-1}$ or less will be quite limited. As facility size increases to 2000 t d^{-1} (the smallest size being considered for most biorefineries), up to 10,000 dt/day, suitable locations will become increasingly limited and/or prices will be considerably higher (Graham et al. 1995).

Facilities with the capability of processing a wide variety of feedstock types may gain a price benefit from seasonal feedstock switching (such as using corn stover in fall, woody crops in winter, and grasses in summer) and from using opportunity feedstock (such as trees blown down from storms). Facilities with a large demand for feedstocks with specific characteristics will depend upon crops genetically selected and tailored for the specific process. The higher feedstock prices would presumably be offset by the economies of scale leading to reductions in processing costs (Jenkins 1997).

Acceptance of biomass facility siting will depend in part on the land area changes required to supply a facility. For economic analysis, it is normally assumed that biomass supplies will be available within an 80 km radius (although in many situations worldwide supply is coming from longer distances, with international trading emerging in some areas). An approximate idea of the hectares and percent land area within an 80 km radius required as a function of the biomass yield that can be sustainably harvested/collected from the land is provided in Table 3.1. The lower yield is representative of levels of agricultural residues (such as corn stover and wheat straw) that could be collected in some areas.

TABLE 3.1 Impact of Facility Size and Energy Crop Productivity on Area Required

Yield t ha ⁻¹ y ⁻¹	Plant Throughput 500 t d ⁻¹		Plant Throughput 2000 t d ⁻¹		Plant Throughput 10,000 t d ⁻¹	
	Hectares	% Area ^a	Hectares	% Area ^a	Hectares	% Area ^a
3.3	44,535	2.2	178,141	8.8	890,705	44.3
9.0	16,330	0.8	65,318	3.3	326,592	16.3
18.0	8,165	0.3	32,659	1.6	163,296	8.13

^a Percent area required within 80-km radius of demand center.

Dedicated energy crops will vary between 9 and 18 t ha⁻¹ y⁻¹ in the near future and could rise higher. Facility demands would be expected to decrease in the future as conversion technology efficiencies increase, requiring less land per unit of final product output. Table 3.2 shows the expected yield of typical secondary energy products from the same daily biomass inputs. The efficiencies are described as *present*, representing the state of the art today, and *future*, in which there are foreseeable changes in the technology.

3.2 Biomass Conversion Technologies

The energy services that can be satisfied by biomass are the supply of heat and combined heat and power, as well as the conversion of biomass into other energy forms such as electricity, liquid, and gaseous fuels identified (see Table 3.2). The transformation of the energy in the raw biomass to other forms of energy and fuels requires an acknowledgment of the physical and chemical differences between biomass resources in order to effect the most efficient transformation at the desired scale of operation. Each conversion process must take into account the differences in biomass physical characteristics on receipt at the processing facility and its proximate composition, which can vary widely in terms of moisture and ash content. Many processes require knowledge of the elemental composition and may even need information at the ultra-structural and polymeric level.

3.2.1 Physical Characteristics

Physical properties are important in the design and handling of biomass fuels and feedstocks at processing plants. In the raw state, woody biomass has a relatively low bulk density compared with fossil fuels. Bituminous coal or crude oil, for example, has a volume of 30 dm³ GJ⁻¹, while solid wood has around 90 dm³ GJ⁻¹. In chip form, the volume increases to 250 dm³ GJ⁻¹ for hardwood species and 350 dm³ GJ⁻¹ for coniferous species. Straw has even less energy density, ranging from 450 dm³ GJ⁻¹ for large round bales to 1–2 m³ GJ⁻¹ for chopped straw, similar to bagasse. The cell walls of biomass fibers

TABLE 3.2 Impact of Facility Size and Process Efficiency on Expected Outputs of Fuels and Electricity

Scale td ⁻¹	Electricity ^a		Ethanol		Fischer–Tropsch Liquids		Hydrogen ^b	
	Capacity MW		Production kL/d		Production Bbl/d		Production t/d	
	Present	Future	Present	Future	Present	Future	Present	Future
500	30.4	45.6	165.9	186.6	453.5	538.5	6.5	7.7
2,000	121.5	182.3	663.5	746.4	1,813.8	2,153.9	25.9	30.8
10,000	607.6	911.5	3,317.5	3,732.2	9,069.0	10,769.5	129.7	154.1

^a Efficiency: present = 30%; future = 45%.

^b Efficiency: present = 50%; future = 60%.

have densities of about 1.5 g cm^{-3} ; however, the large volume of the vascular elements that make up the fibers results in the low energy densities described previously. The fibrous nature also precludes close packing, which is the reason that chips and chopped straw have densities that are so low.

Size reduction of biomass resources is often more difficult than with minerals because the materials are naturally strong fibers, and the production of uniform particle size feedstocks is correspondingly difficult. The size reduction challenge is often mitigated in combustion and gasification on account of the high chemical reactivity of the biomass materials. However, when it is necessary to have penetration of chemicals into the structure of the biomass or to create access for biological agents as in acid and enzymatic hydrolysis of biomass, a higher degree of size reduction will be required. The energy requirements for size reduction are proportional to the surface exposure and rise in inverse proportion to the cube of the average length, so although moderate amounts of energy are required at particle sizes of several millimeters, the energy penalty at the 100- μm size is considerable.

3.2.2 Chemical Characteristics

3.2.2.1 Proximate Analysis

The value of the proximate analysis is that it identifies the fuel value of the as-received biomass material; provides an estimate of the ash handling and water removal requirements; and describes something of the characteristics in burning (Table 3.3). Generally, highly volatile fuels, such as biomass, need to have specialized combustor designs to cope with the rapid evolution of gas. Coals with very high fixed carbon need to be burnt on a grate because they take a long time to burn out if they are not pulverized to a very small size.

The energy content of biomass is always reported for dry material; however, most woody crops are harvested in a green condition with as much as 50% of their mass water. Two different heating value reporting conventions are in use. The term *higher heating value*, or HHV, refers to the energy released in combustion when the water vapor resulting from the combustion is condensed, thus releasing the latent heat of evaporation. Much of the data from North America are reported in this way. The *lower heating value*, or LHV, reports the energy released when the water vapor remains in a gaseous state. The HHV and LHV are the same for pure carbon, which only produces carbon dioxide when burned. For hydrogen, one molecule of water is produced for each molecule of hydrogen, and the HHV is 18.3% greater than the LHV. The HHV is 11.1% greater than the LHV for the combustion of one molecule of methane (CH_4).

3.2.2.2 Ultimate Analysis

The elemental composition of the fuel biomass is usually reported on a totally dry basis and includes the ash. Table 3.4 gives the ultimate analysis for several different fuels.

3.2.2.3 Polymeric Composition

Biomass conversion technologies that depend on the action of microorganisms are very sensitive to the ultrastructure and polymeric composition of biomass feedstocks. In ethanol production and in anaerobic digestion, a complex sequence of biochemical events take place in which the polymers are broken down into smaller units and hydrolysed to give simple sugars, alcohols, and acids that can be processed into fuel

TABLE 3.3 Proximate Analysis of Solid Fuels

Fuel	Ash Content (%)	Moisture (%)	Volatiles (%)	Heating Value (HHV GJ Mg^{-1}) ^a
Anthracite coal	7.83	2.80	1.3	30.90
Bituminous coal	2.72	2.18	33.4	34.50
Sub-bituminous	3.71	18.41	44.3	21.24
Softwood	1.00	20.00	85.0	18.60

^a On a moisture and ash free basis (maf).

TABLE 3.4 Ultimate Analysis Data for Biomass and Selected Solid Fuels

Material	C	H	N	S	O	Ash	HHV (GJ Mg ⁻¹)
Bituminous coal	75.5	5.0	1.20	3.10	4.90	10.3	31.67
Sub bituminous coal	77.9	6.0	1.50	0.60	9.90	4.1	32.87
Charcoal	80.3	3.1	0.20	0.00	11.30	3.4	31.02
Douglas fir	52.3	6.3	0.10	0.00	40.50	0.8	21.00
Douglas fir bark	56.2	5.9	0.00	0.00	36.70	1.2	22.00
Eucalyptus grandis	48.3	5.9	0.15	0.01	45.13	0.4	19.35
Beech	51.6	6.3	0.00	0.00	41.50	0.6	20.30
Sugar cane bagasse	44.8	5.4	0.40	0.01	39.60	9.8	17.33
Wheat straw	43.2	5.0	0.60	0.10	39.40	11.4	17.51
Poplar	51.6	6.3	0.00	0.00	41.50	0.6	20.70
Rice hulls	38.5	5.7	0.50	0.00	39.80	15.5	15.30
Rice straw	39.2	5.1	0.60	0.10	35.80	19.2	15.80

Dry basis, weight percent.

molecules. Wood, grasses, and straws are collectively called lignocellulosics and consist mainly of fibers composed of lignin, cellulose, and hemicellulose. Cellulose, hemicellulose, and lignin are different carbon–hydrogen–oxygen polymers with differing energy contents and chemical reactivities.

Lignocellulosics typically contain cellulose (40%–50%) and hemicellulose (25%–30%) along with lignin (20%–30%), some extractives, and inorganic materials. Cellulose is depolymerized and hydrolyzed to glucose, a six-carbon sugar (C-6), while hemicellulose is a complex mixture of mainly five-carbon sugar (C-5) precursors with xylose, a major product. Lignin is a complex polymer based on phenyl propane monomeric units. Each C₉ monomer is usually substituted with methoxy groups on the aromatic ring, occurring mainly in syringyl (dimethoxy) and guaiacyl (monomethoxy) forms for which the ratio is a signature characteristic of grasses, hardwoods, and softwoods.

Other major plant components containing polymers of carbon, hydrogen, and oxygen that are also used for energy include:

- Starches that are the major part of the cereal grains, as well as the starch from tubers such as mannioc and potatoes
- Lipids produced by oil seed-bearing plants such as soya, rape, or palms, which are possible diesel fuel substitutes when they are esterified with simple alcohols such as methanol and ethanol
- Simple sugars produced by sugar beet in temperate climates or sugar cane in the tropics, which can be directly fermented to alcohols.

In addition to the carbon, hydrogen, and oxygen polymers, there are more complex polymers such as proteins (which can contain sulfur in addition to nitrogen), extractives, and inorganic materials. The inorganic materials range from anions such as chlorine, sulphate, and nitrates, and cations such as potassium, sodium, calcium, and magnesium as major constituents. Many trace elements, including manganese and iron, are the metallic elements in key enzyme pathways involved in cell wall construction.

3.2.3 Combustion Applications

More than 95% of the worldwide use of biomass is through its combustion. The majority is used in developing countries for use in *daily living*, to provide heat for cooking and space heating. In the industrialized countries, a parallel in the cold regions of the northern hemisphere is the *community applications* of district heating. *Industrial applications*—especially of CHP (combined heat and power)—are important in industrialized countries as well as in the agricultural processing sector of developing countries. In several countries, biomass is burned directly to produce electricity without the coproduction of heat; however, like coal and oil-fired steam electricity generation, this is relatively inefficient.

3.2.3.1 Daily Living

Half of all solid biomass combustion worldwide goes into cooking and space heating at the household level, serving the primary fuel needs of over 2 billion people. The typical final energy demand for cooking is around 12 MJ a day for a household in Southeast Asia. This demand can be met with around 20 MJ d⁻¹ of primary energy input using LPG (propane or butane) or kerosene with a typical conversion efficiency of 60%–75%. Biomass efficiencies are much lower, ranging between 10 and 20%; thus, the required primary energy input is between 60 and 120 MJ d⁻¹. The three-stone stove has been extensively studied and has a peak power rating of 5 kW in the startup and a simmer rate of around 2 kW. The measured efficiency with a skilled operator is 10%–15%. Daily household biomass consumption is therefore in the region of 2–4 kg d⁻¹ or about 0.5–1.0 t y⁻¹.

The low efficiency of use is mainly due to poor combustion efficiency and inadequate heat transfer to the cooking vessels. The poor combustion results in the loss of fuel as solid particulate (soot), unburnt hydrocarbons and carbon monoxide; these are often referred to as the products of incomplete combustion (PIC). The heat transfer problem is related to the poor contact between the hot gases from the burner and the pot, and its fouling with PIC (Prasad 1985). The PICs are not only a loss of fuel value, but are also toxic chemicals to which human exposure should be minimized.

Carbon monoxide is a chemical asphyxiant, and the aldehydes, phenols, and other partial combustion products are skin, eye, and lung irritants; some, such as benzopyrenes, are even carcinogens. Because many stoves are used in enclosed spaces and do not have chimneys, the household is exposed to these hazards; eye and lung damage is experienced especially by women and children, who work closely with the stoves while cooking. The epidemiological data on this are extensive (Ezzati and Kammen 2001), and large efforts to improve the efficiency and minimize the emissions in cookstoves are underway. The health impacts are of such a magnitude that subsidized alternative fuels such as LPG could be a significant social benefit while reducing the impacts of biomass harvesting.

3.2.3.2 Space Heating

Space heating with wood in the form of logs is very common in forested areas of the northern hemisphere and Latin America with wood stoves supplying a household heat requirement of 30–100 kW. The thermal efficiencies are about 60%–70%. Log burning is controlled by air starvation to reduce power output and may result in severe emissions of PIC, causing poor air quality in regions with high densities of wood-fueled space heating. Residential wood stoves using advanced combustion system designs or catalysts reduce PIC. Legislation has mandated emissions testing of new appliances before sale to ensure that they meet new source performance standards (NSPS) in the U.S. The use of dry prepared fuels such as pellets (manufactured by drying, grinding, and compression into a pellet about 1–2 cm in diameter) allows combustion to be efficiently managed so that PIC are extremely low with 75%–80% thermal efficiency.

3.2.3.3 Community Systems

Biomass use in community district heating (DH) systems results in improved thermal efficiency with reduced emissions because of the scale of the equipment. Europe has seen large-scale adoption of biomass-fired district heating. There are over 300 round wood-fired DH schemes in Austria with an average heating effect of 1 MW or less, serving villages of 500–3000 inhabitants with minigrids using hot water distribution. Using straw, 54 Danish systems cover larger population centers and range in size from 600 kW to 9 MW. Larger scale DH systems can be CHP units producing process steam for industries, heating for households, and electricity for the grid with high-quality emissions control systems. In Finland, even larger units are serving cities, e.g., the Forssa wood-fired CHP–DH plant is rated at 17.2 MW_e and 48 MW_{th} district heating, generating steam at 6.6 MPa and 510°C, with a bubbling fluidized bed boiler. The seasonal efficiency, which includes a condensing steam cycle in the summer, is 78%.

3.2.3.4 Industrial Applications—Combined Heat and Power (CHP) or Cogeneration

Many industrial processes, such as pulp production or cane sugar manufacture, have large heat requirements that are satisfied by generating low pressure steam at 1.7 MPa and 250° using their own

process residues such as hog fuel (a mix of wood and bark); black liquor (the lignin by-product of pulp manufacture in kraft pulp mills); or bagasse (the fibrous residue after the pressing of sugar cane). A proportion of the steam is used in direct drives for the process plant's needs. Often, because the primary goal has been to eliminate residue disposal, the systems have not operated at maximum efficiency.

In industries such as pulp and paper (P&P), the trend is already toward requiring higher power-heat (P/H) ratios because steam demand is trending down due to energy conservation practices and electricity demand is rising to supply the increasing environmental and process needs. The thermal demand reduction is a consequence of improvements in drying and evaporation technologies, which have decreased from 8.6 to only 4 GJ adt⁻¹ (adt=air dry tonne of pulp). The average pulp mill energy consumption has gone from 17 GJ adt⁻¹ in the 1980s to 7–8 GJ adt⁻¹ in the 1990s.

The application of advanced power generation cycles in the pulp and paper industry using black liquor and wood residues as feedstocks would give large increases in electricity export from that industry. Similar energy conservation trends in the sugar industry can reduce the requirement of steam to process cane from 58.5% (tonne steam/tonne of cane processed) to less than 35%. If investments in energy conservation and adoption of integrated gasification combined cycle (IGCC) technology were applied worldwide, the annual amount of power that could be produced from the large sugar cane resource would be > 650 TWh based on the current > 1.0 Gt of cane. Current cane generation is estimated to be 30–50 TWh y⁻¹ because only sufficient electricity is generated in most instances to power the mill in season.

3.2.4 Electric Power Generation From Biomass

3.2.4.1 Prime Mover Systems and Fuels

Power generation takes place in *prime movers*, a technical term to describe engines of all kinds attached to alternators to generate electricity. Prime movers include the:

- Steam engine
- Steam turbine engine (the Rankine cycle)
- Internal combustion engine (ICE), which comes in two types: the Otto or gasoline-based spark ignition (SI) engine, and the diesel or compression ignition (CI) engine
- Gas turbine engine (Brayton cycle)
- Stirling engine.

Each of these requires that the raw biomass be processed to some level and then used in the prime mover. Eventually fuel cells will replace the prime mover and alternator requirement by generating electricity directly from biomass-derived hydrogen fuels.

The steam cycle (already discussed) uses combustor and boiler combinations to generate steam, requiring that the fuel be reduced in size (perhaps dried to some level) and have physical contaminants removed. The high-pressure steam is then expanded through a steam engine at small scales or through a turbine at larger scales. The efficiency of conversion technologies for combustion steam boiler systems is very scale dependent—a small steam engine or turbine would not exceed 10%. However, typically sufficient biomass is available over a transportation radius of 10–80 km to operate a unit in the range of 10–50 MW electricity output. Current Rankine cycles, at that scale, operate in the range of 25%–30% efficiency and, as a consequence, require approximately 0.75–1 t of dry biomass to produce 1 MWh of electricity.

Industrial and power generation boilers range in size from 100 to 300 MW thermal output. The major types of boilers are: pile burners; grate boilers; suspension fired boilers; fluidized beds; and circulating fluid beds. Recent trends in power generation are to use circulating and bubbling fluidized bed combustors, although the majority of units in current service are stoker-fired moving grate units.

Biomass resources can be used in ICEs and gas turbine systems only if they are converted into clean liquid or gaseous fuels. Ethanol, biodiesel from lipids, or Fischer–Tropsch liquids can be used with little

alteration to SI or CI engines. Gaseous fuels include the mixture of methane and carbon dioxide (biogas) produced by the action of microorganisms on biomass in anaerobic digestion (AD). AD is conducted at an industrial scale, using sewage or effluent streams containing high levels of soluble sugars, alcohols, and acids, as well as in landfills.

The installed landfill power generation capacity in the U.S. is now in excess of 1 GW (Goldstein 2002). The energy content of biogas is typically 20–25 MJ Nm⁻³ or between 50 and 60% that of natural gas. Fuel gases can also be produced by thermal gasification; when this is carried out at small scales, the gasifying agent is usually air. The product gas, which has a low calorific value (LCV) with a heating value of 12%–15% that of natural gas, is often called a producer gas. LCV gas at about 5–6 MJ Nm⁻³ has carbon monoxide (CO), hydrogen (H₂), and methane (CH₄) as the main fuel components, diluted with a lot of nitrogen and carbon dioxide. Larger scale processes can utilize pure oxygen, enriched air, or an indirect heating method to produce medium calorific value (MCV) gas in the range of 15–25 MJ Nm⁻³ heating value with essentially the same fuel gas components but with much less inert diluent. Clean MCV gases can be burnt without much modification of ICEs or gas turbines.

3.2.4.2 Cofiring Biomass With Coal

In a biomass/coal cofiring operation, a biomass fuel is used to replace a portion of the coal fed to an existing coal-fired boiler. This has been practiced, tested, or evaluated for a variety of boiler technologies. There are four types of coal boilers: pulverized coal (PC); cyclone; stoker; and fluidized bed. PC boilers are the most common type in the U. S., representing about 92% of the U.S. coal-generating capacity, with cyclone boilers as the next most common, with about 7% representation. Demonstrations have been undertaken at coal plants ranging in size from about 30 MW_e through to 700 MW_e. Wood residues, hybrid poplar, and switchgrass have all been tested as supplemental fuels, and several utilities have biomass cofiring plants in commercial operation.

Solid biomass can be fed to coal boilers by blending biomass on the coal pile or separately injecting biomass into the boiler. Introducing blended feeds to PC boilers can cause operational problems with the pulverizer, so the biomass proportion is limited to no more than 2%–3% by heat input (4%–5% by mass). Separate injection allows for the introduction of higher biomass percentages to the PC boiler—typically up to 15% on a heat input basis (about 30% by mass).

However, separate injection requires additional fuel handling equipment and increased fuel preparation. Capital costs for the blended feed approach are typically \$50/100 kW⁻¹. For the separate feed approach, capital costs are typically higher, in the range of \$175/200 kW⁻¹. Cofiring can reduce boiler efficiency to some degree. For example, cofiring in a PC boiler with 10% of the heat input from wood may decrease the boiler efficiency by 0.5%–1.5%; after “tuning” the boiler’s combustion output and adjusting for any efficiency losses, the combustion efficiency to electricity would be approximately 33%.

Because coal plants comprise more than half of U.S. power plant capacity currently in operation, cofiring technology has the advantage of improving environmental performance at existing power plants, while providing fuel source flexibility and using proven and familiar equipment. It reduces air pollution emissions, GHG emissions, and the amount of waste ash generated as a by-product of the combustion. In addition, it requires relatively low up-front capital expenses compared to other renewable energy options; this makes it a straightforward and inexpensive way to diversify the fuel supply and divert biomass from landfill disposal.

3.2.5 Thermal Gasification Technologies

The conversion of biomass into a gaseous fuel opens up modern applications in electric power generation, the manufacture of liquid fuels, and the production of chemicals from biomass. The chemistry of gasification of biomass is best viewed as an extension of the pyrolysis process. Pyrolysis is simply defined as the chemical changes occurring in the solid biomass when heat is applied to a material in the absence of oxygen. The products of biomass pyrolysis include water, charcoal (or more correctly

char, a carbonaceous solid), oils or tars, and permanent gases including methane, hydrogen, carbon monoxide, and carbon dioxide.

The majority of gasifiers are partial oxidation reactors, in which just sufficient air or oxygen is introduced to burn part of the input biomass to provide the heat for pyrolysis and gasification. If the oxidant is air, the product gas is diluted by the nitrogen present. Although air is 79% nitrogen, the stoichiometry of partial oxidation is such that the final LCV product, gas, has about 50% nitrogen as a diluent. The energy content of the typical gases produced in biomass gasification is shown in Table 3.5. The use of pure oxygen as the gasification agent eliminates the nitrogen diluent and can produce medium calorific value (MCV) gases in the range of 10–20 MJ Nm⁻³.

An alternative strategy is to carry out the gasification process by means of indirect heat the product stream is even higher in calorific value, because neither nitrogen nor the carbon dioxide produced from the combustion in situ of the partial oxidation processes is present in the product gas stream. The challenges to achieve a clean and useable fuel gas have been addressed through gasifier design and postgasification processing to remove tar and particulate contaminants from the gas stream.

3.2.5.1 Gasifier Systems

The main challenge in gasification is enabling the pyrolysis and gas-reforming reactions to take place, using the minimum amount of energy, in reactors that are economical to construct. During a history dating back to the late 18th century, an extraordinary number of different designs and process configurations have been proposed. Prior to the development of fluidized bed technologies in the 1920s, the majority of the gasifiers were so-called fixed bed units. The flow of gasifying agents, usually air and steam, could be cocurrent with the biomass feed, or countercurrent; these are often described, respectively, as downdraft and updraft gasifiers.

Downdraft gasification was widely used during the Second World War as an on-board fuel gas generator to offset the lack of gasoline. Millions of units were constructed and then abandoned as soon as petroleum supplies resumed. Units derived from the automotive application are marketed today as stationary generating sets equipped with ICEs, with SI or CI for power production in remote locations and in developing countries without grid systems.

The simplest and oldest gasifier is the counterflow moving bed, which consists of an insulated shaft into which the feedstock (typically pieces larger than 3 cm on a side) are fed. The shaft is filled to a depth of 0.5–2 times the vessel diameter, and the mass of material is supported on a grate. The oxidant (air or enriched air/oxygen) and sometimes steam are introduced below the grate. The grate material is ignited and the hot gases flow up through the bed, exchanging heat with the down-flowing biomass material and, at the same time, pyrolyzing and drying it. At steady state, the bed level is maintained by continuous additions of the feed.

TABLE 3.5 Heating Values of Fuel Gases

	HHV ^a Btu ft ⁻³ ^c	LHV ^b	HHV ^a MJ Nm ⁻³ ^d	LHV ^b
Hydrogen	325	275	12.75	10.79
Carbon monoxide	322	322	12.63	12.63
Methane	1013	913	39.74	35.81
Ethane	1792	1641	69.63	63.74
Propane	2590	2385	99.02	91.16
Butane	3370	3113	128.39	118.56

Note: Conversion factors for 1 MJ Nm⁻³ at 273.15 K and 101.325 kPa. 25.45 Btu ft⁻³ at 60°F and 14.73 psia. Inverse 1 Btu ft⁻³ at 60°F and 30 in. Hg. 0.0393 MJ Nm⁻³.

^a Higher heating value.

^b Lower heating value.

^c Standard temperatures and pressure of dry gas are 60°F and 14.73 psia (NIST 2004).

^d S.I. units.

The product gases include the desired fuel gases (methane, carbon monoxide, hydrogen, and C_2 hydrocarbons), nitrogen, carbon dioxide, and water vapor, which exit at relatively low temperatures ($< 250^\circ\text{C}$). The water arises from the moisture content of the feed and the products of combustion. The gas also contains several percent of condensable tars. Because the temperatures in the combustion zone can be very high, problems with the formation of slags (molten mixtures that can impede the oxidant flow) may occur. Some of the systems are constructed deliberately to form a molten ash or slag, and this is handled by dropping the slag into a receiving tank, where it solidifies and shatters to an easily handled aggregate.

The throughput of air-based systems is relatively low, handling about $150\text{ kg m}^{-2}\text{ h}^{-1}$ of dry biomass. This operational limitation is due to the need to reduce the rate of heat generation at the grate to avoid slugging in dry systems. The lower throughput rate also helps to manage the problems of maintaining a uniform flow of the descending material to avoid channeling and even greater reductions in throughput. The largest units are typically 3–4 m in diameter and handle between 50 and 100 t d^{-1} of material with a rating of 10–20 MW_{th} input. The hot gas efficiency can be as high as 80% conversion of the input energy into a fuel gas and sensible heat. The cold gas efficiency will be about 70%.

If the gas is maintained at temperature and then fired in a kiln or boiler, the presence of tars and particulate are of little consequence because the tars are combusted in the excess air and the particulate is removed in the kiln exhaust or boiler flue gas system. Such close-coupled systems are quite common applications of biomass gasification in several industries and the basis of district heating systems used in Finland. The gas typically has a heating value of 6 MJ Nm^{-3} on a dry basis and is richer in methane, ethene, and ethyne than most other gasifier systems using air. For engine applications the gas must be cleaned up and cooled to ambient temperatures. Typically, this is done with some sort of aqueous quench and scrubbing system (e.g., a wet electrostatic precipitator) to remove tar and particulate matter, and the tar/water removed must be separated and then treated for release to the environment.

Fluidized beds can remove the grate limitation of the moving bed gasifier because, due to their high mixing rates and outstanding heat transfer rates, they can achieve through-puts of over $1500\text{ kg m}^{-2}\text{ h}^{-1}$ of dry biomass. The two modes of fluidized bed operation involve bubbling (BFB) and circulating (CFB) solids. In a BFB, the inert fluid bed medium (silica sand, dolomite, alumina, and olivine) is retained in the body of the bed, which is expanded and in motion (literally bubbling) due to the flow of the fluidizing gas medium into the bed. The terminal velocity of a particle in the flow is still higher than the linear flow rate of the gas and thus remains in the body of the unit. In the CFB, the gas flow is increased to the point that the terminal velocity of the particle is exceeded and it is transferred out of the expanded bed and captured in a cyclone for return to cycle.

The effective rating of a 3-m diameter unit is over 50 MW_{th} input of biomass. Because the combustion process and the pyrolysis process are now mixed, the exit gas temperatures are typically 700°C – 800°C . Due to cracking processes, the tar production is lower and mainly composed of secondary and tertiary tar components; however, the tolerance of high moisture content feedstocks is much more limited than the countercurrent moving bed unit. The system design is also more complex, requiring blowers to inject the oxidant at the base of the fluidized bed. In addition to the expansion of the gas due to temperature, the gas volume increases due to the formation of the fuel gas. This necessitates careful design of the CFB and the freeboard in a BFB; in a CFB, this is accommodated by the fact that the entire bed is in circulation. The oxidant requirement is typically $0.3\text{ kg O}_2\text{ kg}^{-1}$ of dry feedstock.

The highest reactivity gasification systems are entrained flow units in which the feedstock is finely divided and burnt in a substoichiometric flame at high temperatures. The postcombustion gases are then allowed to reach chemical equilibrium before being quenched. Such units produce very few higher hydrocarbons and tar materials. The challenge for biomass is to obtain a sufficient rate of fuel injection to the flame. One configuration is the Koppers–Totzek gasifier. For fossil fuels, the Texaco gasifier uses a dense slurry of coal or petroleum coke in water as the feed to a pressurized oxygen flame, producing a hydrogen and carbon monoxide mixture for use as a syngas or as a fuel in gas turbines.

Concurrent moving bed gasifiers (also known as downdraft and cross-draft units) were widespread during World War II (various estimates put the number at over 1 million). Like the other gasifiers

described earlier, the design is a partial oxidation system. The physical arrangement is a shaft reactor into which the biomass is introduced at the top; the material then flows down to an oxidation zone, where the air is admitted. This zone feeds heat back to the incoming biomass to pyrolyze the material into gases, tars, and charcoal. The charcoal is partially combusted in the oxidation zone, and the products of pyrolysis flow through the incandescent combustion zone to be cracked into, primarily, hydrogen and carbon monoxide as the fuel gases, with few or no tars passing through. The combusted gases flow on through the hot charcoal, reacting with it, until the rate of reaction effectively goes to zero because the reactions are very endothermic.

The net result is a gas at about 500°C that is very low in condensible tars; other than carbon dioxide, water vapor, and nitrogen, the fuel gases are mainly hydrogen and carbon monoxide with only a small amount of methane and higher hydrocarbons. The low tar production is the reason this was the favored system for on-road vehicles in the 1940s. Like all moving bed reactors, the operation depends on the presence of an adequate void space between particles. The wood feedstock must produce a char that has sufficient strength to support the column of feedstock without crushing and blocking the flow.

The overall efficiency is lower than the countercurrent system for two reasons. Chemically, it is not possible to convert the charcoal fully; the sensible heat from the combustion zone is not transferred to the incoming feed material, but rather must be heat exchanged with air oxidant prior to being quenched. Because of the need for uniformity of mass and heat transfer in the high-temperature reaction zone, the diameter of such units is generally limited to less than 1.5 m and, with typical throughputs of $300 \text{ kg m}^{-2} \text{ h}^{-1}$ of dry biomass, the size of units ranges from 100 kW_{th} to 2 MW_{th} input. The chemistry of this gasifier has a lot in common with iron blast furnaces and is significantly different from the other gasifiers, which have very high burdens of liquids such as tars in the product gas.

Medium calorific value (MCV) gas can be produced without using oxygen or air in gasification. This requires an independent source of heat to be transferred to the biomass to pyrolyze the feedstock under conditions of high severity, i.e., long residence times at high temperatures. To this end, proposals have been made to use concentrated solar energy or high-temperature nuclear reactors as the heat source in the gasification of a wide range of feedstocks. In the case of biomass, the pyrolysis gasification process will provide a char stream with slow or fast pyrolysis to temperatures of 600°C–750°C, in amounts representing between 12 and 25% of the input biomass. This contains sufficient energy to drive the pyrolysis process.

A number of developments in which biomass is pyrolyzed using a solid heat carrier in an inert atmosphere or steam or product MCV gas to generate the MCV fuel gas and char are already in the demonstration phase. The char and the cooled heat carrier, which can be silica sand, alumina, or a mineral such as olivine, are then transferred to a separate air combustion unit in which the char is combusted to heat up the heat carrier, which is then returned to the pyrolysis gasification unit. Mechanical separation and isolation devices in the heat carrier circulation loops prevent the passage of fuel gas from the gasifier to the oxidation conditions of the combustor. The transportation mechanism is via fluidization of the heat carrier.

To this end, a group in Austria constructed a version with a CFB combustion unit and BFB pyrolysis gasifier; in the U.S., units have been constructed with a CFB pyrolysis reactor and a BFB combustor as well as a dual CFB configuration scaled up to 60 MW_{th} biomass input. Another variant, developed by Manufacturing and Technology Conversion International, Inc. (MTCI), a U.S. company, was a BFB pyrolysis and steam char reaction system with the heat supply from immersed heat exchanger bundles in the bed. The heat exchanger bundles are fueled with part of the product gas, and the char from pyrolysis is consumed, as in the cocurrent gasifier.

3.2.5.2 Commercialization

Use of gasifiers as precombustors in which the hot gases are passed through a high-temperature duct into a boiler or industrial process kiln has been widespread; cofiring is an example. Foster Wheeler Energia Oy CFB has been supplying a LCV gas for several years to an existing large-scale utility boiler at the Kymijärvi 167 MW_e and 240 MW_{th} fossil-fired plant close to Lahti, Finland. A combination of forestry residues,

industrial wood residues, and recycled fuel (a clean fraction of the municipal waste stream composed mainly of wood, paper, and cardboard separated at the source) is the fuel source. This project builds on many years of successful operation of biomass CFBs in thermal applications and substitutes for about 15% of the total fuel used in the boiler.

Applications in which clean gases (free of particulate and tar) are utilized occur mainly in the development and demonstration stages, with the exception of deployment of cocurrent gasifier–ICE generator sets. In 2000, Sydkraft A.B. successfully completed a high-pressure LCV gas biomass integrated gasification and combined cycle (IGCC) operation, which was part of the Värnamo (Sweden) district heating system. This first complete biomass-fueled pressurized CFB IGCC was fueled with about 18 MW_t equivalent of wood residues and produced about 6 MW_e (4 from the gas turbine, 2 from the steam cycle) and 9 MW_{th}. The LCV gas is cooled to 350°C–400°C before it is cleaned in metal filters and then passed to the Typhoon gas turbine, which is manufactured by Alstom. The project demonstrated more than 1500 h of IGCC operation on the product gas prior to its shutdown at the end of the demonstration phase (Stahl et al. 2000).

Small-scale gasifiers for the production of fuel gas for cooking, as well as the demonstration of electricity in ICEs, are already commercial in India and China. Their introduction has been accelerated by the use of moderate government subsidies, especially for communities not served by the electricity grid. In development circles, such systems are often referred to as village power systems. Considerable discussion among the OECD nations concerns the use of distributed energy resources, which would provide local power and heat (e.g., CHP units) while remaining connected to the grid. Many of these units are based on natural gas; however, when the economics and availability of biomass are appropriate, gasification is considered; demonstrations of gasifier–ICEs with CHP applications in district heating have taken place in Denmark.

3.2.6 Biological Process Technologies

The polymers that make up biomass are produced through photosynthetically driven biological reactions at the cellular level in the plant. The resulting phytomass is then fuel and food (or feedstock) for a large number of the fauna on Earth. From simple bacteria to large mammals, the processes to digest and metabolize plant polymers have evolved over billions of years. In a number of instances, the biological degradation processes have been harnessed in industrial conversion processes to produce biofuels and bioproducts to serve humanity. Of the many processes, only two have attained large-scale use: anaerobic digestion and ethanol production.

3.2.6.1 Anaerobic Digestion

Anaerobic microorganisms convert the biodegradable fraction of organic materials to carbon dioxide and methane in the absence of oxygen. Commercial-scale anaerobic digestion has been used in many parts of the world for many years and is only beginning to be commercial in the U.S. Several utilities are currently producing “green energy” with biogas from landfills. Biogas from anaerobic digesters on dairy and hog farms and cattle feed lots is beginning to contribute to the energy mix in the U.S.

3.2.6.2 Ethanol Production

Ethanol is currently the most important liquid fuel produced from biomass. It is produced by fermentation of sugars. The fermentable sugars can be obtained directly from sugar cane and sugar beet, or they can be produced by means of the hydrolysis of starches or cellulose and hemicelluloses. Starch crops include cereals, and manioc in tropical countries, while cellulose and hemicellulose are the major polymers of wood, grasses, straws, and stalks. The production from sugar cane in Brazil has been in the range of 11–13 GL y⁻¹ for the last few years; in the U.S., the annual production has grown rapidly from 5.42 GL in 2000 to 8.08 GL in 2002. This growth is primarily due to legislation banning the use of methyl tertiary-butyl ether as an octane enhancer in several state markets.

Over 90% of the U.S. production is from corn, which is processed in wet or dry mills. Corn kernels have about 70%–75% starch content; approximately 10% protein (gluten); 4%–5% germ (the source of corn oil); and 3%–4% fiber content. Dry milling consists of grinding the whole kernel into a flour and then slurring this with enzymes to break down the starch to fermentable sugars. The stillage separated from the beer is composed of coarse grain and a wide range of soluble materials. The solubles are concentrated by evaporation into a syrup (condensed distillers solubles), which is then added to the coarse grain to produce dried distillers grain with solubles or DDGS. This is marketed as a feed for animals. Dry milling theoretically produces about 460 L of ethanol per tonne of corn.

Wet milling is an example of a biomass refinery. The corn kernels are steeped in dilute sulfurous acid for 48 h in a mild prehydrolysis process prior to a grinding step to separate the corn germ. Corn oil is recovered from the germ as high-value cooking oil. The starch and the gluten protein are then separated. The gluten is dried and sold as a 60% protein feed to mainly poultry markets. The starch fraction is the source of dried or modified cornstarches and a syrup converted enzymatically into high-fructose corn syrup or passed to a fermentation step producing ethanol and other fermentation-derived chemicals.

Yields of ethanol are somewhat lower than those from dry milling at around 440 L t⁻¹ of corn. According to the National Corn Growers Association (www.ncga.com), corn utilization rates of the 246.25 Mt, 2002 harvest (9.695 billion bushel) were: 58% as feed; 19% as export grain; 6% as high-fructose corn syrup; 9% as fuel ethanol; and 8% from all other applications (including breakfast cereal, cornstarches, and seed corn).

Since the inception of the corn-to-ethanol process technology in the 1970s, continuous improvement has occurred in terms of yield, process efficiency, reduced capital investment, and nonfeed operating costs. The industry average yield has gone from 360 L t⁻¹ (2.4 USgal/bu) to 410 L t⁻¹ (2.75 USgal/bu), while the capital cost has declined from 220 to about 115\$ L⁻¹ d⁻¹ capacity (2.5 down to 1.3 USD per annual gallon of capacity), according to the industry.

3.2.7 Liquid Fuels and Bioproducts From Lignocellulosics

Lignocellulosics (such as wood, straw, and grasses) contain cellulose (40%–50%) and hemicellulose (25%–30%). There is considerable ethanol potential through fermentation of the sugars derived from these polymers, producing similar theoretical yields per tonne as corn, i.e., 450 L t⁻¹ and a price structure, which is more stable relative to the fuel markets than foodstuffs. Alternative processes would include thermochemical routes through the production of synthesis gas and its catalytic synthesis to mixed alcohols, Fischer–Tropsch liquids, or hydrogen—a future transportation fuel (Table 3.2).

Current RD&D in the U.S. is focused on the development of two product platforms: a sugar platform that inherits much of the prior research into the production of ethanol from lignocellulosic biomass, and a syngas platform evolved from the development of biomass gasifiers primarily for power production coupled with the extensive downstream catalysis developments of the petrochemical industry. The syngas platform is also the basis for the combustion units for the utilities component of the platforms, which are shown diagrammatically in Figure 3.1.

The sugar platform inherits over two decades of work on the ethanol conversion process from lignocellulosics (Wyman 1994). The polymers involved are much more recalcitrant than starches and, because of the complex nature of lignocellulosics, they require extensive effort to break down the lignin, cellulose, and hemicellulose structure so that the individual polymers become available for hydrolysis. Cellulose is hydrolyzed to glucose, a six-carbon sugar (C-6), while hemicellulose is a complex mixture of mainly five-carbon sugar (C-5) precursors, with xylose as a major product. Although the C-6 sugars are relatively easy to ferment with yeasts, such as *Saccharomyces* spp., the C-5 sugars have not been as easy to ferment to ethanol (Zhang et al. 1995). The pretreatment stages can include steam, acid, and alkali treatments, while the hydrolysis steps can be carried out with acids or enzymes. Because of inhibition of the enzymatic hydrolysis by the sugars produced, the National Renewable Energy Laboratory (NREL) developed a simultaneous saccharification and fermentation process to remove the sugars as they are

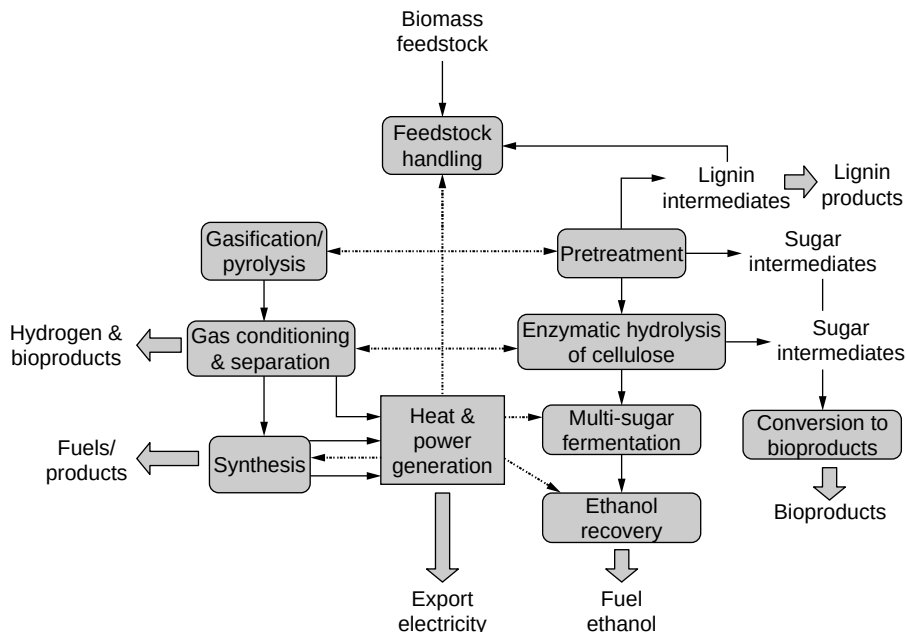


FIGURE 3.1 The U.S. RD&D approach to the sugar and syngas platforms.

formed by producing ethanol in the same reactor. Such process integration will be the key to producing low-cost sugars and thus ethanol or other bioproducts in the future (Lynd et al. 1999).

The syngas platform using biomass would enable the production of renewable liquid fuels and chemicals such as ammonia and hydrogen by well-established and proven syngas technologies. Today these processes utilize the syngas produced by the steam reforming of natural gas or use syngas from coal gasification. The production of liquid fuels from syngas has a long history dating from the pioneering work of Fischer and Tropsch to synthesize hydrocarbon fuels in Germany in the 1920s. The process was implemented in Germany in the 1930s where nine plants operated during World War II on syngas from coal gasification; after the war, until the mid 1950s, they operated on syngas from the gasification of heavy oil. In the 1950s, South Africa constructed the first of the South African Synthetic Oil Limited (SASOL) plants, an activity that has continued its development to the present day using syngas produced from coal and natural gas. Both of these developments were initiated during times of petroleum shortages, to fuel the transportation systems of Germany and South Africa.

Continuing research and development is based on the environmental advantages of producing zero sulfur fuels, with combustion properties tailored to specific engine designs to minimize emissions from petroleum-derived fuels. At the time of writing, there are at least six developers of what are known as Fischer–Tropsch liquids (FTL) plants, which are scaled to use remote and difficult-to-get-to-market sources of natural gas as the feedstock for the syngas. FTL processes use catalysts based mainly on iron, cobalt, ruthenium, and potassium and have been extensively characterized. They operate at high pressures between 2.5 and 4.5 MPa, and temperatures between 220 and 450°C. The reactors range from fixed catalyst beds to bubbling as well as circulating fluidized beds. The product distribution depends on the polymerization on the surface of the catalyst. The hydrocarbon chains are built up one carbon atom at a time—a process known as homologation. Polymer growth of this type is now understood to take place via what is known as the Anderson–Shulz–Flory mechanism, and it is not possible to produce a single compound with a fixed number of carbon atoms or even a narrowly defined distribution of molecular weights.

Extensive product recycling and recirculation of unreacted syngas can be used to reduce the number of small hydrocarbon molecules (e.g., C_1 to C_5); however, production of wax-like molecules with as many as 30–50 carbon atoms also exists. The conversion per pass of the syngas over the catalysts is often limited, due in part to the fact that the polymerization reaction is exothermic and the equilibrium mixtures obtained are very temperature sensitive, affecting the yield and proportions of the desired products. The recycle processes are very energy intensive and reduce the throughput of the catalyst system.

Recent progress in enhancing the yields of gasoline and diesel products has been made partly by optimization of the catalysts and operating conditions, and also by companies such as Shell with its Shell Middle Distillate Synthesis (SMDS) process in which the higher molecular weight waxes are hydrotreated over selective catalysts to increase the yield of the desired range of hydrocarbons C_8 through C_{20} . The smaller carbon number products, C_1 to C_5 , are also reformed to increase the hydrogen to the overall process. Other innovations in the FTL arena include slurry reactors, which use very finely divided catalysts suspended in an oil medium. Such reactors often demonstrate much increased conversion in a single pass and offer economies with respect to recycle energy and investment. However, from the chemical engineering perspective; such three-phase reaction systems have required a lot of R&D and development.

An alternative production process would use a high conversion single-pass catalyst such as a slurry reactor and then utilize the tail gas after the catalyst as fuel for a gas turbine combined cycle. Such a process would then coproduce electricity, heat, and a liquid fuel and offer lower investment costs and higher overall efficiency.

Higher alcohols are produced from syngas over typical Fischer–Tropsch catalysts under conditions of greater severity than those used predominantly in the production of hydrocarbons. Pressures are between 5 and 15 MPa and are generally at temperatures higher than FTL synthesis. The Anderson–Shulz–Flory mechanism does not appear to determine the product distribution, although homologation through the addition of one carbon atom at a time is followed, so that product methanol is recycled over the catalyst to grow into higher alcohols.

Changes in catalyst composition can determine the isomer composition of the higher ($>C_3$) alcohols. Also, mixed thermochemical syngas and biotechnology routes to ethanol are under development. Ethanol can be synthesized from syngas using an anaerobic bacterium, *Clostridium ljungdahlii*. The growth conditions of the bacterium are managed so as to maximize the ethanol yield over the production of acetate (as acetic acid) from the key biological pathway intermediate acetyl-CoA, which is derived from the syngas. Although not yet commercialized, many of the development challenges of producing the bacterium and making effective gas–liquid contactors and bioreactors have been overcome. Yields are similar to those from the inorganic high temperature Fischer–Tropsch catalysts.

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For Further Information

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Additional information on ethanol can be obtained from the Web sites of the National Corn Growers Association (<http://www.ncga.com>) and the Renewable Fuels Association (<http://www.ethanolrfa.org/>).

The last three decades of biomass activity in the United States are described in: Chum, H. L. and Overend, R. P. 2003. *Biomass and Bioenergy in the United States*. In *Advances in Solar Energy: an Annual Review of Research and Development*. Y. Goswami, ed. American Solar Energy Society, Boulder, CO. U.S.A. 83–148.

Additional information on energy efficiency and renewable energy technologies can be obtained from the U.S. Energy Efficiency and Renewable Energy Websites <http://www.eere.energy.gov/biomass.html>, and <http://www.eere.energy.gov/RE/bioenergy.html>. Also the Website <http://bioenergy.ornl.gov/> provides useful resource information and many links to other bioenergy Websites.